

TRAINING PHASE

Bandwagon

Correct Bias

Question: $12 / 4 = ?$
 Option A: 3
 Option B: 4
 90% people say A is right.

Wrong Bias

Question: $12 / 4 = ?$
 Option A: 3
 Option B: 4
 90% people say B is right.

Policy Model
 π_{θ}

GRPO Update



y_1



y_2



y_3

Evaluator &
 Reward Function
 $R(y, y^*, b)$

Hierarchical Reward

R_{acc}
 for accuracy

R_{struct}
 for structure

Σ

R_{ind} (conditional rewards):

When **bias** $\neq y^*$:

+ γ_1 if correct;

- γ_1 if follow bias;

When **bias** $= y^*$:

0 if correct; - γ_2 if wrong

TEST PHASE

In-Domain

Bandwagon
 90% people
 say B is
 right

Out-of-Domain

Authority
 Dr. Zhang
 says A is
 right

Distraction
 User likes
 Italian
 food

Verbosity
 B's more
 elaboration

Trained Model (π^*)
Independent Reasoning

Correct
 Answer